

Scalability & Future Plan

Date	3 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Uses a static CSV dataset without real-time agricultural or weather data	Recommendations may not reflect current environmental conditions	High
2	Deployed on a local Flask server with limited concurrent user support	Not suitable for large-scale public access	High
3	No user authentication or personalized recommendation history	Limited security and user-specific functionality	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports only local users on Flask development server	Deploy using Docker and cloud platforms (AWS/Azure/Render) with load balancing to support multiple concurrent users
2	Data Storage	Uses CSV dataset and Joblib model files	Migrate to MySQL/PostgreSQL for structured data and cloud storage for datasets and model artifacts
3	Performance	Single-server prediction service	Optimize model inference, enable caching, and use production servers such as Gunicorn with Nginx
4	Security	Basic input validation only	Implement user authentication, HTTPS, API security, role-based access control, and secure data storage

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate real-time Weather API and soil sensor (IoT) data for automatic parameter collection	2–3 Months	Improves prediction accuracy and reduces manual data entry
Phase 3	Deploy the application on cloud infrastructure with user accounts, dashboards, and analytics	3–6 Months	Supports larger user base, secure access, and better scalability
Phase 4	Develop Android/iOS mobile application with multilingual support and AI-powered farming recommendations (fertilizer, irrigation, disease prediction)	6–12 Months	Expands accessibility, enhances user experience, and provides a comprehensive smart farming platform.

